

Common approaches and tips.

- Identify the enabling systems and materials needed for transition early in the life cycle to allow for the necessary lead time to obtain or access them.

Elaboration

This section elaborates and provides “how-to” information on the Stakeholder Needs and Requirements Definition process. Further guidance on elicitation, life cycle concepts, needs and requirements definition can be found in ISO/IEC/IEEE 29148 (2018), the INCOSE GtWR (2023), the INCOSE GtNR (2022), and the INCOSE NRM (2022).

Verified and validated stakeholder needs and stakeholder requirements are drivers and constraints for the majority of the system life cycle Technical Processes. Depending on the system development model, life cycle concepts definition, and maturation, the stakeholder needs and stakeholder requirements capture should be conducted at the beginning of the development cycle and assessed as a continuous, concurrent, and iterative activity as the project team moves recursively through the system architecture and across all life cycle activities. The reason for eliciting and analyzing the life cycle concepts, stakeholder needs, and stakeholder requirements is to understand the expectations of stakeholders well enough to support the System Requirements Definition processes.

Identify Stakeholders. One of the biggest challenges in system development is the identification of the set of stakeholders from whom life cycle concepts, needs, and requirements are elicited. When identifying stakeholders, take into account those who may be affected by, are able to influence, or will support the life cycle stages of the SoI. Typically, stakeholders include customers, users, operators, maintainers, procurement, organization decision makers, approving authorities, regulatory bodies, developing organizations, verifiers, validators, support organizations, and society at large (within the context of the business and

proposed solution). This can include the stakeholders of external systems (e.g., interoperating, interfacing, other constituent systems in a system of systems) and enabling systems, as these will usually impose constraints that need to be identified and considered in the SoI or could have impacts on those systems or the environment. In sustainable development, this includes identifying representation for future generations. When direct contact is not possible, agents are identified, such as marketing or user groups to represent the concerns of classes of stakeholders such as consumers or future generations. There also may be stakeholders who oppose the system. These detractors of the system are first considered in establishing consensus needs. Beyond this, they are addressed through the Risk Management process, the threat analysis of the system, or the system requirements for security, adaptability, agility, or resilience.

Elicit or Derive Stakeholder Needs and Stakeholder Requirements.

Determining stakeholder needs and requirements requires the integration of a number of disparate views, which may not necessarily be harmonious. It is important to have a “reconcile” path in the establishment of stakeholder needs and stakeholder requirements, since the stakeholder expectations and the life cycle concepts may be in conflict, incomplete, ambiguous, infeasible, or unable to be satisfied collectively within project constraints. This circumstance illustrates an aspect of “horizontal integration” (see [Section 2.3.5.8](#)), recognizing that there will often be prioritization of competing concerns, or even outright rejection of some stakeholder concerns because of inconsistencies with other stakeholders’ needs and requirements or a lack of feasibility.

As the SE processes are applied, a common paradigm for examining and prioritizing available information and determining the value of added information should be created. Each of the stakeholder’s views of the needed systems can be translated to a common system description that is understood by all participants, and all decision-making activities recorded for future examination. The stakeholder views will be influenced by cognitive biases (see [Section 1.4.2](#)) based on their specific role, education, work experiences, culture, etc. Stakeholder views are framed in the context

of these biases. It is important for the project team to understand this during elicitation to better understand the perspective of each stakeholder.

SE practitioners support project management in defining what must be done and gathering the information, personnel, and analysis tools to elaborate the life cycle concepts, needs, and requirements. This includes eliciting or deriving stakeholder needs, stakeholder requirements, system/project constraints (e.g., cost and schedule constraints, technology limitations, applicable specifications, and requirements), “drivers” (e.g., capabilities of the competition, external threats, and critical environments), and risks.

The output of the Stakeholder Needs and Requirements Definition process should be sufficient definition of the life cycle concepts, stakeholder needs, and stakeholder requirements to gain authorization and continuing funding for through the Portfolio Management process (see [Section 2.3.3.3](#)). The output should also provide necessary technical definition to the Acquisition process (see [Section 2.3.2.1](#)) to generate a request for supply if the system is to be acquired through an acquisition or to gain authorization to develop and market the system if the SoI is to be developed within the organization.

Since stakeholder needs and requirements come from multiple sources, eliciting and capturing them constitutes a significant effort on the part of the project. The life cycle concepts help the project team understand the context within which the needs and requirements are captured and defined. Modeling, analysis, and simulation tools can also be used to evaluate candidate solutions and select a desired solution (see [Section 3.2.1](#)).

It is essential to establish a database of the data and information which represents the artifacts generated during this process. The database also includes traces between the stakeholder needs, stakeholder requirements, and system requirements. They serve as a foundation for later refinement

and/or revision by subsequent activities across the life cycle. Tools for capturing and managing requirements can be used.

Refine Life Cycle Concepts. Stakeholder needs and requirements result from obtaining an understanding of stakeholder expectations through the definition, analysis, and maturation of in a series of life cycle concepts (e.g., acquisition concept, deployment concept, operations concept, support concept, and retirement concept). Development of preliminary life cycle concepts were introduced in the Business or Mission Analysis process (see [Section 2.3.5.1](#)). These life cycle concepts need to be refined as part of the Stakeholder Needs and Requirements Definition process.

The primary objective of the development of life cycle concepts is to ensure that stakeholder needs and requirements are clearly understood and the rationale for each is incorporated into the decision mechanism for later transformation into the system requirements. Interviews with manufacturing/coding stakeholders, operators, maintainers, and disposers of current/similar systems, potential users, owners of interoperating, interfacing, and enabling systems (see [Section 1.3.3](#)), and site visits provide valuable stakeholder input toward establishing life cycle concepts. Other objectives are as follows:

- To provide traceability between stakeholder needs and stakeholder requirements and their source.
- To establish a holistic understanding of the capabilities needed to address the problem or opportunity in terms of people, process, and products.
- To establish a basis for needs and requirements to support the system over its life, such as personnel requirements, enabling systems, and support requirements.
- To establish a basis for design, system verification, and system validation planning across the life cycle and resulting artifacts and requirements for enabling systems needed as part of the validation and verification activities.

- To assess interactions of the SoI with users and its operating environment including interactions across interface boundaries with external and enabling systems.
- To provide the basis for analysis of system performance, behavior under (over)-load, and mission-effectiveness calculations.
- To validate needs and requirements at all levels and to discover implicit requirements overlooked from other sources.

The life cycle concepts are used to define an integrated set of stakeholder needs which are transformed into the set of stakeholder requirements.

Uncertainties and Risk. During the development of the preliminary life cycle concepts as part of the Business or Mission Analysis process (see [Section 2.3.4.3](#)), there may have been uncertainties (see [Section 1.4.1](#)) from several perspectives including business, market, management, technical performance, schedule, development and production costs, operations and support costs, security, and sustainability. These uncertainties are a source of risk. Each of these uncertainties must be addressed during the Stakeholder Needs and Requirements Definition process and further elaborated during the Systems Requirements Definition Process (see [Section 2.3.4.3](#)).

Record and manage the life cycle concepts, needs, and requirements. The life cycle concepts, stakeholder needs, and stakeholder requirements should be recorded and managed within the project database in a form that allows traceability between the life cycle concepts and the resulting stakeholder needs and requirements (see the INCOSE GtNR [2022] and the INCOSE NRM [2022] for more details).

2.3.5.3 System Requirements Definition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.3.1] The purpose of the System Requirements Definition process is to transform the stakeholder, user-oriented view of desired capabilities into a technical view of a solution that meets the operational needs of the user.

Description. System requirements are the foundation of system definition and form the basis for the System Architecture Definition, Design Definition, Integration, and Verification processes. Each requirement carries a cost, so the system requirements should be the minimum set necessary and sufficient to realize the intent of the stakeholder needs and requirements. Typically, the later in the project that changes are introduced to the system requirements, the greater the impact is to cost and schedule. Where there is more uncertainty in the requirements, the uncertainty should be managed until the requirements mature.

The System Requirements Definition process generates system requirements from a technical perspective using the stakeholder needs and requirements that reflect the stakeholders' perspectives. As such, the stakeholder needs and requirements drive and constrain the SoI being developed. The quality of the resulting system requirements is dependent on the quality of the agreed-to stakeholder needs and requirements.

System requirements definition is concurrent, iterative, and recursive. Thus, the System Requirements Definition process is done concurrently and iteratively with the other Technical Processes, particularly the Stakeholder Needs and Requirements Definition and the System Architecture Definition processes. With each iteration, more detailed information is discovered and defined based on the analysis and maturation of the life cycle concepts and the system solution. In addition, the System Requirements Definition processes is recursively applied to define the requirements for each lower-level system element within the SoI architecture. The allocation of the system requirements is performed concurrently with the System Architecture Definition process. Lower-level system elements are defined via the System Architecture Definition process, and then the SoI level requirements are allocated to the system elements at the next level. For

each lower-level system element, the Stakeholder Needs and Requirements Definition and System Requirements Definition processes are repeated recursively until all system elements have their system requirements defined. The outputs of System Requirements Definition process must be traceable and consistent with the life cycle concepts and stakeholder needs and stakeholder requirements, without introducing unnecessary implementation biases. The System Requirements Definition process adds the verification criteria to each system requirement as it is derived.

Inputs/Outputs. Inputs and outputs for the System Requirements Definition process are listed in the IPO diagram in [Figure 2.41](#). Descriptions of each input and output are provided in Appendix E.

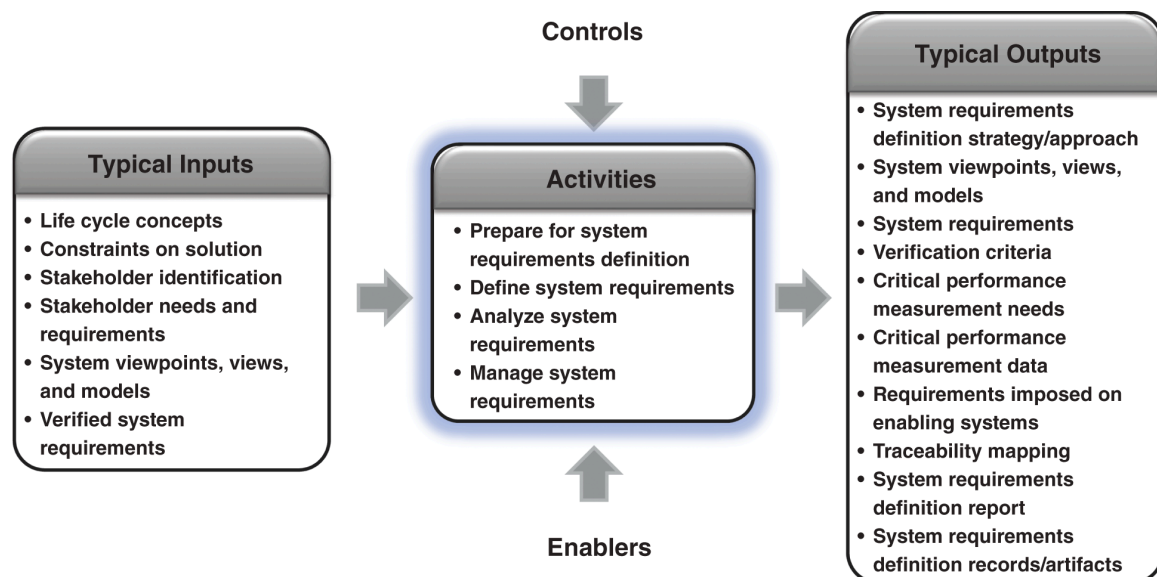


FIGURE 2.41 IPO diagram for System Requirements Definition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities. The System Requirements Definition process includes the following activities:

- *Prepare for system requirements definition.*
 - – Establish the strategy/approach for system requirements definition.
 - – Plan for the necessary enabling systems or services needed through the life cycle for system requirements definition. Enabling systems include tools for elicitation of requirements, life cycle concepts, recording drivers and constraints, defining risks, analysis, recording

system needs, recording system requirements, and providing traceability.

- – Ensure enabling system or service access needed to support system requirements definition.
- *Define system requirements.*
 - – Define the functional boundary of the system in terms of the behavior and properties to be provided.
 - – Identify the life cycle concepts and stakeholder requirements from which the system requirements will be transformed and then define each function and associated performance.
 - – Define each expected system function, including the associated performance. Include both primary functions and enabling functions.
 - – Define necessary constraints. These include higher-level requirements allocated to the SoI, operational conditions, and interactions with external systems. Define interactions with users, operators, maintainers, and disposers.
 - – Identify system requirements that relate to risks, criticality of the system, critical quality characteristics, and compliance with standards and regulations.
 - – Define verification success criteria for each system requirement, the verification strategy, verification method, and responsible organization for providing proof the system requirements have been met (see [Section 2.3.5.9](#)).
 - Capture the system requirements and their attributes.
- *Analyze system requirements.*
 - – Analyze the system requirements for characteristics of individual requirements and of the set of requirements (can be the set of requirements for the current increment, build, or sprint). Analyze the set of requirements to ensure they are correct, complete, consistent, comprehensible, appropriate to level, and feasible (see the elaboration below and the INCOSE GtWR [2023] for more details).
 - – Enable technical achievement monitoring through the definition of critical performance measures.

- – Review the analyzed requirements with the applicable stakeholders.
- – Perform issue resolution for the system requirements. Negotiate changes, amendments, and modifications to resolve inconsistencies, conflicts, and unrealizable or impractical requirements.
- *Manage system requirements.*
 - – Confirm agreement that the system requirements meet the stakeholder needs and requirements.
 - – Capture key system requirements decisions, rationale, alternatives, and enablers.
 - – Establish and sustain traceability (system requirements).
 - – Manage system requirements change.
 - – Give CM the information items, work products, or other artifacts needed for baselines.

Common approaches and tips.

- Identify the enabling systems and materials needed for transition early in the life cycle to allow for the necessary lead time to obtain or access them.

Elaboration

This section elaborates and provides “how-to” information on the System Requirements Definition process. Additional guidance on needs and requirements definition can be found in ISO/IEC/IEEE 29148 (2018), the INCOSE GtWR (2023), the INCOSE GtNR (2022) and the INCOSE NRM (2022).

Stakeholder Requirements versus System Requirements. The set of stakeholder requirements are SoI requirements written from the stakeholders’ perspectives to represent what they require of the SoI in order to meet their needs. The set of system requirements represent the technical perspective of what the SoI must meet during the System Architecture Definition and Design Definition processes that will result in a system that meets the stakeholder needs and stakeholder requirements.

Another key distinction is that the focus of the stakeholder requirements is often on high-priority and critical functions, performance, quality, compliance, etc., while the system requirements are more encompassing and detailed including enabling functions, performance, quality, compliance, etc. that will result in the stakeholder requirements to be implemented.

In some cases, the stakeholder requirements can be copied directly into the set of system requirements “as is” and additional requirements added as needed. For smaller, internal projects, the set of stakeholder requirements could be used as the set of system requirements, depending on how much analysis went into the definition of the set of stakeholder requirements such that their implementation will result in the stakeholder needs to be met.

When a set of stakeholder requirements is provided to a supplier by an acquirer, the supplier uses these as inputs to their SE processes to develop the set of system requirements. When defining the system requirements, rather than treating the supplied stakeholder requirements as the *only* source of requirements, the supplier has an obligation to do an assessment for derived system requirements and as well as requirements from other “non-acquirer” stakeholders. For example, the supplier’s production team needs the product to be manufacturable, their test team needs the product to be testable, the supplier and public need the product to be safe and secure from a cybersecurity perspective, the users and operators need the product to be easy and safe to interface with from a human perspective, and the organization has regulatory compliance considerations. For products to be developed by an outside supplier, the supplier’s company may need the product to conform to a strategic development effort aligning with other products produced, internal standards, and technology maturation. The acquirer may not have included all of these considerations when developing their set of requirements. If the supplier blindly follows only the acquirer supplied requirements specified in their contract, they are likely to generate a SoI that may not work in the integrated system or operational environment, resulting in a system that fails system validation.

Plan for system requirements definition. The System Requirements Definition process should begin with a review of the problem, threat, or opportunity for which the SoI is to address, and the mission, goals, objectives and critical success measures defined by the Business or Mission Analysis process (see [Section 2.3.5.1](#)) and the set of stakeholder needs, stakeholder requirements, and life cycle concepts defined by the Stakeholder Needs and Requirement Definition process (see [Section 2.3.5.2](#)). For contracted development efforts, mission, goals, objectives, and critical success measures can come from both the acquirer and supplier organizations. Before the System Requirements Definition process, the project team will need to define the strategy to be used to transform the stakeholder needs and stakeholder requirements, define drivers and constraints, assess risks, define, analyze, and mature life cycle concepts, and derive an integrated set of system requirements resulting from these activities.

Requirements Definition. The integrated set of stakeholder requirements is transformed into system requirements to address what the system must do to meet those needs. The transformation process involves additional analysis and further elaboration of the models developed during life cycle concept analysis and maturation. The system requirements must address function, fit, form, quality, and compliance with stakeholder and business needs. System requirements must also address interactions with external systems, users, operators, maintainers, disposers, and the operational environment. SE practitioners collaborate with the stakeholders of the external systems to define each of the interactions and record an agreement of those definitions in some configuration managed form, as well as any constraints or interface requirements (see [Section 3.2.4](#)).

Definition of the system requirements is a complex process that includes function and performance analysis; trade studies; constraint evaluation; inclusion of (or reference to) specific requirements from relevant standards and regulations; risk assessment; technology assessment; detailed characterization of the operational environment; detailed assessment of the interactions of the parts that make up the SoI, detailed assessment of

the interactions between the SoI and users, operators, maintains, disposers, and external systems; and cost–benefit analysis. System requirements cannot be established without determining their impact (achievability) on lower-level system elements, especially in terms of cost, schedule, and technology. Therefore, system requirements definition is a concurrent, iterative, and recursive balancing process that works both “top-down” (called allocation, derivation, and flow-down) and “bottom-up” (called compliance analysis and flow-up).

The system requirements are inputs to the System Architecture Definition and Design Definition processes, in some domains these requirements are referred to as “design-to” or “design input” requirements. When the requirements are defined, it is important that they are expressed at a level of abstraction that is appropriate to the SoI and systems hierarchy level to which they apply. Although it is good practice to avoid implementation when defining the system requirements, it is not always possible.

In defining system requirements, care should be exercised to ensure each requirement statement is appropriately crafted. The characteristics shown in [Table 2.7](#) should be considered for each individual requirement statement (INCOSE GtWR, 2023). In addition to the characteristics of individual requirement statements, the characteristics shown in [Table 2.8](#) should be considered for requirement sets (INCOSE GtWR, 2023).

TABLE 2.7 Requirement statement characteristics

Requirement Statement Characteristic	Definition
Necessary	The requirement statement defines a capability, characteristic, constraint, or quality factor needed to satisfy a life cycle concept, need, source, or parent requirement.
Appropriate	The specific intent and amount of detail of the requirement statement is appropriate to the level (e.g., the level of abstraction, organization, or system architecture) of the entity to which it refers.
Unambiguous	The requirement statement is stated such that the intent is clear and the requirement can be interpreted in only one way by all the intended stakeholders.
Complete	The requirement statement sufficiently describes the necessary capability, characteristic, constraint, conditions, or quality factor to meet the need, source, or higher-level requirement from which it was transformed.
Singular	The requirement statement states a single capability, characteristic, constraint, or quality factor.
Feasible	The requirement statement can be realized within entity constraints (e.g., cost, schedule,

Requirement**Statement****Characteristic****Definition**

technical, legal, ethical, safety) with acceptable risk.

Verifiable

The requirement statement is structured and worded such that its realization can be verified to the approving authority's satisfaction.

Correct

The requirement statement is an accurate representation of the need, source, or higher-level requirement from which it was transformed.

Conforming

The requirement statement conforms to an approved standard pattern and style guide or standard for writing and managing requirements.

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TABLE 2.8 Requirement set characteristics

Requirement Set Characteristic	Definition
Complete	The requirement set for a given SOI should stand alone such that it sufficiently describes the necessary capabilities, characteristics, functionality, performance, drivers, constraints, conditions, interactions, standards, regulations, and/or quality characteristics without requiring other sets of requirements at the appropriate level of abstraction.
Consistent	The requirement set contains individual requirements that are unique, do not conflict with or overlap with others in the set, and the units and measurement systems they use are homogeneous. The language used within the sets is consistent (i.e., the same words are used throughout the set to mean the same thing). All terms used within the requirement statements are consistent with the architectural model, project glossary, and project data dictionary.
Feasible	The requirement set can be realized within entity constraints (e.g., cost, schedule, technical) with acceptable risk.
Comprehensible	The requirement set is written such that it is clear as to what is expected of the entity and its relation to the macro system of which it is a part.

Requirement Set Characteristic	Definition
Able to be validated	The requirement set will lead to the achievement of the set of needs and higher-level requirements within the constraints (such as cost, schedule, technical, and regulatory compliance) with acceptable risk.
Correct	The requirement set is an accurate representation of the needs, sources, or higher-level requirements from which it was transformed.

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System requirement statements may have a number of attributes attached to them (either as fields in a database or through relationships with other artifacts) shown in [Table 2.9](#). *The attributes annotated with an asterisk (“*”) represent a proposed minimum set.* See the INCOSE NRM (2022) for the definition and description of these attributes.

TABLE 2.9 Requirement attributes

Attributes to Help Define Needs and Requirement and Their Intent

A1—Rationale*

A2—Trace to Parent*

A3—Trace to Source*

A4—States and Modes

A5—Allocation/Budgeting*

Attributes Associated with System Verification and System Validation

A6—System Verification or System Validation Success Criteria*

A7—System Verification or System Validation Strategy*

A8—System Verification or System Validation Method*

A9—System Verification or System Validation Responsible Organization*

A10—System Verification or System Validation Level

A11—System Verification or System Validation Phase

A12—Condition of Use

A13—System Verification or System Validation Results

A14—System Verification or System Validation Status

Attributes to Help Manage the Requirements

A15—Unique Identifier*

A16—Unique Name

A17—Originator/Author*

A18—Date Requirement Entered

A19—Owner*

A20—Stakeholders

A21—Change Control Board

A22—Change Proposed

A23—Version Number

A24—Approval Date

A25—Date of Last Change

A26—Stability/Volatility

A27—Responsible Person

A28—Need or Requirement Verification Status*

A29—Need or Requirement Validation Status*

A30—Status of the Need or Requirement

A31—Status (of Implementation)

A32—Trace to Interface Definition

A33—Trace to Dependent Peer Requirements*

A34—Priority*

A35—Criticality or Essentiality*

A36—Risk (of Implementation)*

A37—Risk (Mitigation)*

A38—Key Driving Need or Requirement

A39—Additional Comments

A40—Type/Category

Attributes to Show Applicability and Enable Reuse

A41—Applicability

A42—Region

A43—Country

A44—State/Province

A45—Market Segment

A46—Business Unit

Attributes to Aid in Product Line Management

A47—Product Line

A48—Product Line Common Needs and Requirements

A49—Product Line Variant Needs and Requirements

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Allocation, derivation, and flow-down. The next level of the system hierarchy is defined in conjunction with the System Architecture Definition and Design Definition processes. System requirements are allocated to the system elements at the next level of the system hierarchy. Once the allocation has been determined, the system requirements are derived (assigned) for the next system elements at the level of system hierarchy such that the intent of the allocated parent requirement is met.

The System Requirements Definition process is repeated recursively for each level of the system hierarchy until the system elements are to the level of detail needed to be realized via a make (e.g., build, code), buy, or reuse decision. The resulting sets of system requirements for the system elements represent the allocated baseline of the SoI.

Requirements Management. According to ISO/IEC/IEEE 29148, requirements management encompasses those tasks that record and maintain the evolving requirements and associated context and historical information from the requirements engineering activities. Effective requirements management occurs within the context of an organization's project and Technical Processes. Requirements management also establishes procedures for defining, controlling, and publishing the baseline

requirements for all levels of the SoI. The resulting sets of requirements are provided to the Configuration Management process (see [Section 2.3.4.5](#)) process for baselining at the appropriate time. The Configuration Management process is used to establish and maintain configuration items and baselines. Requirements management also ensures traceability is established between requirements and other artifacts (see [Section 3.2.3](#)), that appropriate requirements reviews occur, and requirements measures are established and used. See also the INCOSE GtNR (2022) and the INCOSE NRM (2022) for further elaboration concerning requirements management.

2.3.5.4 System Architecture Definition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.4.1] The purpose of the System Architecture Definition process is to generate system architecture alternatives, select one or more alternative(s) that address stakeholder concerns and system requirements, and express this in consistent views and models.

System Architecture Definition process transforms related architectures (e.g., strategic, enterprise, reference, and SoS architectures), organizational and project policies and directives, life cycle concepts and constraints, stakeholder concerns and requirements, and system requirements and constraints into the fundamental concepts and properties of the system and the governing principles for evolution of the system and its related life cycle processes. This process results in a system architecture description for use by the project, its organization, other organizations, and various stakeholders. The Project Management Plan (PMP) and Systems Engineering Management Plan (SEMP) in some cases will provide management directives on how to perform this process, but usually the programmatic view and other related views developed by the System Architecture Definition activities will guide the PMP and the SEM. The architecture governance activities at the organization level will provide

additional direction for the System Architecture Definition process through its issuance of architecture governance directives. Since the directives and stakeholder requirements can evolve throughout the system life cycle, the system architecture description should be treated as a living artifact reflecting both the changing expectations and the evolution of our understanding of what the system solution should be.

Development practices for architecture are specified by ISO 15704 for enterprises and the ISO/IEC/IEEE 42000 series of standards in software, systems, and enterprise fields of application. ISO 15704 specifies terms, concepts, and principles considered necessary to address stakeholder concerns, carry out enterprise creation programs and any incremental change projects required by the enterprise throughout its whole life. ISO/IEC/IEEE 42000 series of standards establishes processes, key principles, and concepts for conceptualization, evaluation, and description of architectures.

Description The System Architecture Definition process provides information and data useful and necessary for identifying and characterizing the fundamental concepts and properties of the system and its elements. These concepts and properties can be fundamentally human-centric, with individual, social, organizational, and political aspects, in human activity systems considering technical elements as enablement assets. The architecture information and data will be implementable through system and system element designs, which satisfy as far as possible the problem or opportunity expressed by models and views for a set of stakeholder and system requirements (traceable to business/mission requirements, as applicable) and life cycle concepts (e.g., Operational, Acquisition, Deployment, Support, and Retirement). During a stage in the system life cycle, the relevant enabling systems and the SoI are considered together as a solution but are distinguished from each other in the overall solution conceptualization.

System architecture definition focuses on achieving associated missions and characterizing the operational concepts of the system and performing

market analysis to ensure viability of the SoI. It utilizes architectural principles and concepts to define the high-level structure of a system and its elements, and the intended properties and characteristics of the SoI. It highlights and supports trade-offs for the other System Definition processes, and possibly Portfolio Management and Project Planning. It incorporates incremental insights obtained about the emergent properties and behaviors of the SoI while achieving a balance for suitability, viability, effectiveness, and affordability. This process is iterative and requires participation of architects, SE practitioners, and specialists in relevant domains, subject matter experts and other stakeholders. The process continues recursively through the levels of the system and its system elements, with consistent feedback to ensure the system design continues to satisfy stakeholder needs and system requirements,

Inputs/Outputs Inputs and outputs for the System Architecture Definition process are listed in [Figure 2.42](#). Descriptions of each input and output are provided in Appendix E.

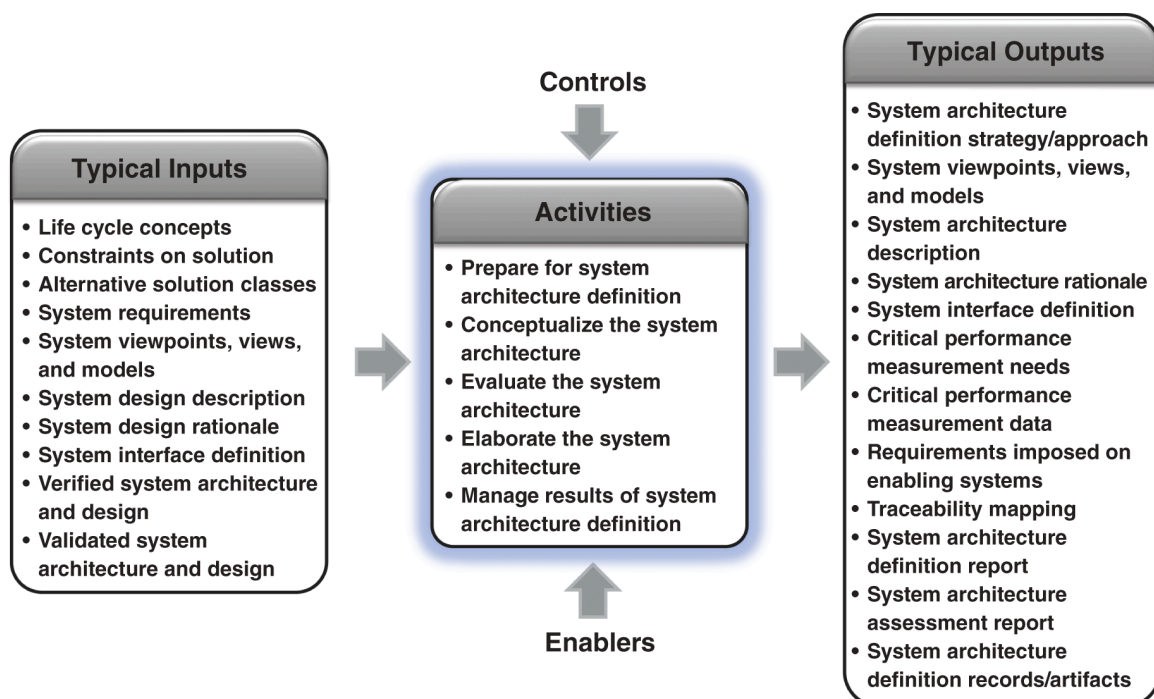


FIGURE 2.42 IPO diagram for System Architecture Definition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The System Architecture Definition process includes the following activities:

Prepare for system architecture definition.

- Identify and analyze relevant market, industry, stakeholder, organizational, business, operations, mission, legal, and other information and related perspectives that will guide the development of the system architecture.
- Identify key milestones and decisions to be informed by the system architecture effort. In particular, identify those key architecture artifacts and resources that guide the system architecture development.
- In conjunction with the System Requirements Definition process, determine the system context (i.e., how the SoI fits into the external environment) and system boundary are refined, that reflect operational scenarios and expected system behaviors. This task includes identification of expected interactions of the SoI with system elements, or other systems or entities.
- Establish the approach for architecting. This includes an architecture roadmap and strategy, methods, frameworks (see [Section 3.2.5](#)), patterns (see [Section 3.2.6](#)), modeling techniques, tools, and the need for any enabling systems (see [Section 1.3.3](#)), products, or services. The approach should also include the process requirements (e.g., measurement approach and methods), evaluation (e.g., reviews and criteria), and necessary coordination.
- Ensure the enabling items (registry, repository, library, competencies), services, resources and capabilities for executing the System Architecture Definition process are available. This includes planning for the need and identifying the requirements for the enabling items.
- Establish or identify associated architecture viewpoints and model kinds that facilitate analysis and understanding of the viewpoint. This task includes identifying expected uses and users of architecture information, identifying potential architecture framework(s), capturing rationale for selection of viewpoints, templates, metamodels and model kinds, selecting, modifying and developing relevant methods, techniques, and tools.

Conceptualize the system architecture.

Note: This activity is based on the Architecture Conceptualization process in ISO/IEC/IEEE 42020

- Characterize the problem space in conjunction with the BMA process and document it. The report focuses on architecture considerations that span one or more system life cycle stages
- In conjunction with the SNRD process, the system context and system boundary are refined, including identification of expected interactions of the SoI with system elements, or other systems or entities. This task includes determination of boundary conditions, quality measures, situation contexts, assumptions, degrees of freedom, constraints, conditions, and challenges.
- Define architecture objectives and critical success criteria that will be used to assess the extent to which the problems and opportunities will be addressed.
- Based on existing or previous solutions, and problem mitigation strategies, address the highest priority requirements and architecture considerations to synthesize a set of potential solutions. This task includes scanning for relevant technologies, problem patterns, solution patterns, naturally occurring solutions, enhancements to existing systems, heuristics, tactics, and discussion with experts.
- For each potential solution, identify strengths, weaknesses, gaps or shortfalls, required trade-offs, consequences, obligations, assumptions, critical success factors affecting critical success criteria and key performance indicators. Devise structural, behavioral, organizational and architectural entities (functions, input/output flows and flow items, states and modes, functional and physical interfaces, nodes and links, computational and communication resources, etc.) to formulate candidate architecture(s). Based on the set of candidate architecture(s), select the best architecture(s) for downstream use by using the Decision Management and Risk Management process. This task includes identifying and characterizing tradeoffs, defining context and scope, determining and mitigating risks, and identifying issues and areas for improvement.

- Select, adapt, or develop views and models of the best architecture(s), by capturing concepts, properties, decisions, processes, activities, tasks, characteristics, guidelines, and principles and utilizing architecture viewpoints to develop architecture descriptions. This task includes determining the scope, breadth and depth, use and users of each view and model, and expressing them in the specified form with sufficient level of detail.

Evaluate the system architecture.

Note: This activity is based on the Architecture Evaluation process in ISO/IEC/IEEE 42020.

- Determine evaluation objectives and criteria for value assessment and architecture analysis by identifying relevant mandates and imperatives, stakeholders and their concerns, policies and standards, value, and quality characteristics.
- Determine evaluation methods and integrate them with evaluation objectives and criteria.
- Collect and review evaluation related information including views and models, architecture concepts, properties, metrics and measures, sources of information, accuracy, errors, degrees of uncertainty, and qualification of correctness, completeness, and consistency of gathered information.
- Analyze, assess, and characterize architecture(s), by using evaluation methods and criteria, and applying the System Analysis and Measurement processes to produce architecture assessments. Architecture alternatives that are similar to each other or fail to meet identified mandates are eliminated and costs, risks, and opportunities are identified and characterized for appropriate actions.
- Formulate, capture, validate, and communicate the findings and recommendations, including implications, to relevant decision makers and stakeholders. The combined overall evaluation can be used to select a preferred system architecture solution.

Elaborate the system architecture.

Note: This activity is based on the Architecture Elaboration process in ISO/IEC/IEEE 42020.

- Based on the identified viewpoints, develop architecture models and views that adequately address stakeholder concerns, while, if applicable, conforming to selected architecture frameworks.
- Perform preliminary interface definition for interfaces with the level of detail necessary for understanding the architecture for decision making and risk management. The definition includes the internal interfaces between the system elements and the external interfaces with entities outside the system boundary.
- Analyze the architecture models and views for consistency and resolve any issues identified. ISO/IEC/IEEE 42010 correspondence rules from frameworks can aid in this analysis. This task includes relating architectural entities to elements of views and models, mapping related entities to relevant architecture and system concepts, properties, and principles, and assessing whether architecture views are consistent with corresponding viewpoints.
- In conjunction with the Verification and Validation processes (see [Sections 2.3.5.9](#) and [2.3.5.11](#)), verify and validate the models by execution or simulation, if modeling techniques and tools permit, and with traceability matrix of operational concepts. Where possible, use design tools to check their feasibility and validity. As needed, implement partial mock-ups or prototypes, or use executable architecture prototypes or simulators.
- Utilizing models and views, develop architecture descriptions by composing those views and models that adequately cover the uses and users of the architecture descriptions. Assess the architecture description against the intent of the architecture, as well as its suitability, correctness, completeness, and consistency.

Manage results of system architecture definition.

- Capture, maintain, and manage the rationale for selections among alternatives and decisions about the architecture, architecture framework(s), viewpoints, model kinds, views, and models. This task includes managing information for decisions, risks, constraints and assumptions and possible governance of upper-level architectures.
- Establish the means for the implementation of the directives of the governance of the architecture, including the roles, responsibilities, authorities, and other control functions. Monitor and assess whether governance directives and guidance are being followed.
- Establish a means for management of the architecture, including plans, measures, schedules, milestones, and other functional outcomes. Monitor and control the implementation of management instructions, provision of status reports, and corrective actions.
- Manage the maintenance and evolution of the architecture, including the architectural entities, their characteristics, and principles. Allocation and traceability matrices are useful to analyze impacts on the architecture.
- Manage the architecting effectiveness, including work performance tracking, reviewing, regulating the progress, dealing with management issues, dealing with resource allocation issues, dealing with methods and tools availability, and coordinating review of the architecture to achieve stakeholder agreement.
- Maintain bi-directional traceability of the system architecture including traceability between the architectural entities to the requirements, interface definitions, analysis results, related architectures, and stakeholder concerns.
- Manage the maintenance, evolution, and use of the architecture descriptions, including the architecture viewpoints, views, and models.

Common approaches and tips.

- Define the problem and the solution spaces with regard to the identified stakeholders
- Define the main principles governing the whole life cycle processes of a SoI, in the scope of the solution space.

- Identify the enabling systems and materials needed for transition early in the life cycle to allow for the necessary lead time to obtain or access them.
- Ensure that conflicting interests (e.g., performance vs. quality characteristics, distributed control vs. central control, new technologies vs. COTS) have been properly addressed.
- Use the Risk Management process to help ensure that the inherent risks associated with the use of new technologies are adequately assessed.

Elaboration

Architecture Processes ISO/IEC/IEEE 42020 (2019) provides a generic process reference model for architecture processes for enterprise, system, and software levels. The concept of architecture as considered in this standard is applicable for different kinds of entities being architected. It specifies 6 architecture processes for use by organizations and projects. As shown in [Figure 2.43](#), the core architecture processes as outlined in the standard are: Architecture Conceptualization, Architecture Evaluation, and Architecture Elaboration. The Architecture Conceptualization process characterizes the problem space and determines suitable solutions that address stakeholder concerns, achieve architecture objectives, and meet relevant requirements. The Architecture Evaluation process determines the extent to which one or more architectures meet their objectives, address stakeholder concerns, and meet relevant requirements. The Architecture Elaboration process describes or documents an architecture in a sufficiently complete and correct manner for the intended uses of the architecture.

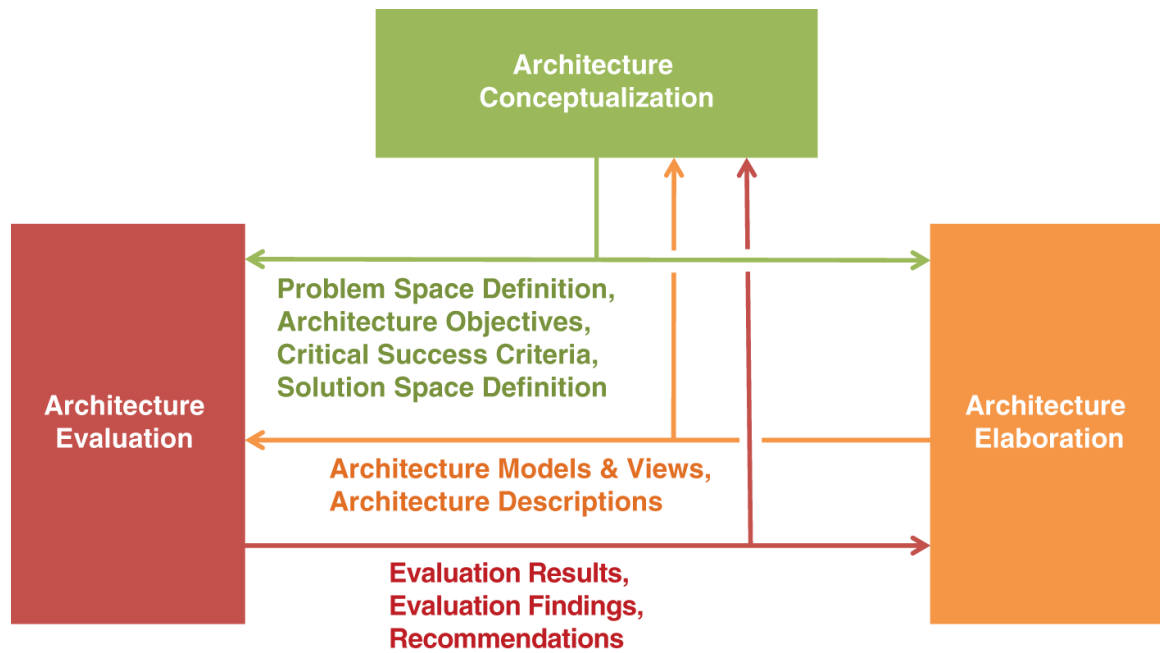


FIGURE 2.43 Core architecture processes. From ISO/IEC/IEEE 42020 (2019). Used with permission. All other rights reserved.

System Architecture The notion of a system is abstract, but it is a practical means to create, design, or redesign products, services, or enterprises. The SoI and the enabling systems that are necessary for development, utilization and support should be considered together in a solution to address a problem or an opportunity. Note that there may be several potential solutions to address the same problem or opportunity. System is represented with sets of interrelated entities—including human in socio-technical systems—achieving one or more stated purposes. These system entities may possess characteristics such as dimensions, environmental resilience, availability, robustness, learnability, execution efficiency, openness, modularity, scalability, and mission effectiveness.

Architecture Description ISO/IEC/IEEE 42010 specifies the normative features of architecture frameworks, architecture description languages, and viewpoints and views as they pertain to architecture description. An architecture description expresses the architecture of a system and is composed of architecture views. A view is an information part comprising portion of an architecture description. It is composed of view components which are derived from models and non-model sources of information. A viewpoint is the set of conventions for the creation, interpretation and use of views to frame one or more concerns of stakeholders and specifies the

ways in which the view components should be generated and used. An architecture framework contains standardized viewpoints, view templates, metamodels, model templates, etc. that aid in development of architecture views. An architecture description language contains syntax and semantics intended for describing the architecture and provides a way to create and understand view components.

Architecture Evaluation ISO/IEC/IEEE 42030 provides a generic, conceptual guiding framework that can be used for the planning, execution, and documentation of architecture evaluations. The elements described in this standard can be used to determine architecture value, determine architectural characteristics, validate whether the architecture addresses current and future stakeholder needs with architecture assessment against defined stakeholder acceptance criteria, and also provide inputs to decisions made at the business, operational and tactical levels.

Architecture Considerations Per ISO/IEC/IEEE 42010, Stakeholder concerns, architecture aspects and stakeholder perspectives are kinds of architecture considerations. Architecture frameworks help identify views and viewpoints to characterize the architectures with regard to these considerations.

Kinds of Architecture Entities Architecture is increasingly applied to systems and other entities that are not traditionally considered to be systems, such as enterprises, services, business functions, mission areas, product lines, families of systems, and software items. Corresponding to each of these entities, different kinds of architecture can be considered according to their purpose, domains of application, and roles within entity and architecture life cycles.

System Architecture vs System Design The System Architecture Definition process focuses on the essential concepts, properties' structure, behaviors, and features that apply to the system solution. It helps gain insights into the relation between the requirements for the system and the emergent properties and behaviors of the system that arise from the interactions and

relations between the system elements. The Design Definition process focusses on developing an overall system design that is ultimately sufficiently detailed to allow its realization. An effective architecture is as design-agnostic as possible to allow for maximum flexibility in the design trade space. The Design Definition process provides feedback to the System Architecture Definition process to consolidate or confirm the allocation, partitioning, and alignment of architectural entities to system elements that comprise the system.

Architecting Styles Per (Evans, 2014), Architecting Styles provide a set of proven approaches for those who create, commission, use, and evaluate architecture products. These can help key decision makers to be better informed on the use and limitations of the architecture thereby ensuring that the different architecting activities consistently deliver value. These styles help to understand the architecting approach; architecture objectives; architectural entities; how value is created to make effective architecture-related decisions. The styles are driven by the purpose, culture, or reason for the architecture and reflect currently observed good practices. The four styles of architecting are: authoritative, directive, coordinative, and supportive.

Architecture Styles Per (Garlan, et al., 1994, 1996), an architecture style is a set of design elements or principles or properties or a generic pattern that provides guidance for the System Architecture Definition process. The set helps in identification and classification of architectures. Architecture styles can be understood as language, system of types and as theory. Architecture styles can be defined by architecture views, architectural elements and their relationships, architecture viewpoints, layouts, connections, interfaces, interaction mechanisms, communication factors, and applicable constraints.

Architecture Patterns Per (Bass, et al., 2012), an architecture pattern is a reusable, configurable architectural entity comprising a minimal set of elements that is complete under certain aspects and exhibits rules for instantiation that is applicable for different situations. It solves and

delineates certain elements of the system architecture and can be used in many system architecture efforts. It has a fundamental structure of predefined elements and relationships, principles, rules and guidelines. Architecture patterns promote communication, streamline documentation, support high levels of reuse, improve architect's efficiency and productivity, and provide a starting point for additional ideas.

Value and Quality While the Systems Architecture Definition process creates a framework for addressing stakeholder concerns and requirements; the goal is to deliver value to all stakeholders, which might correlate to quality factors deemed important. It is essential that value is created over the life of the system so that the system remains satisfactorily in use. The perception of what is of value to stakeholder changes over time, and hence it is necessary to account for the different times at which value is being presented or reported. This requires that sources of stakeholder value are determined, system capabilities are defined to produce or influence value, and vulnerabilities that cause value degradation are identified. Per Kumar (2020), value-based approaches helps one to learn and understand the stakeholder's value system, their principles of behavior, expectations, ideals and belief systems, motivation, and the boundaries within which the stakeholder can be engaged.

Notion of Interface The notion of interface is one of the key items to consider when defining the architecture of a system. The term "interface" comes from Latin words "inter" and "facere" and means "to *do* something *between* things." Therefore, the fundamental aspects of an interface are functional and defined as inputs and outputs of functions. Interoperability is a stakeholder need and requirement, ensuring interfaces use open, well maintained and enduring standards is key to reduce future integration challenges.

Horizontal and vertical integration System Architectures ensure that requirements allocated throughout the system's design process account for system elements and interfaces as the design matures. The architecture establishes the significant operational and system development interfaces,

both internal and external, that must be maintained through development and upgrades. The overall System Architecture is composed of system elements, which are integrated to form the entire system. It is essential to maintain cognizance of the end-to-end system performance expectations when evaluating integration of the system elements, so that those elements continue to perform as needed. When a dynamic relationship exists between one element in the system and another, there is an interdependency. This may involve relationships that are functional or physical in nature, or both. Depending on how tightly coupled these system elements are, the net effect on the system will vary. For example, there is often an interdependency between safety functional hazard conditions and certain function and physical system elements defined in the system architecture.

2.3.5.5 Design Definition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.5.1] The purpose of the Design Definition process is to provide sufficient detailed data and information about the system and its elements to realize the solution in accordance with the system requirements and architecture.

This process is driven by requirements that have been vetted through the architecture and more detailed analyses of feasibility.

Description The Design Definition process transforms architecture and requirements into a design of the system that can be realized. This process results in sufficiently detailed data and information about the system and its elements to enable implementation consistent with architectural entities defined in models and views of the system architecture, in conformance with applicable system requirements, and in alignment with design guidelines and standards adopted by the organization or project. Often

these system elements are identified, and their fundamental concepts and properties are characterized, by the System Architecture Definition process. The design information and data will define the expected properties and characteristics allocated to each system element and enable transition toward their realization.

Inputs and Outputs Inputs and outputs for the Design Definition process are listed in [Figure 2.44](#). Descriptions of each input and output are provided in Appendix E.

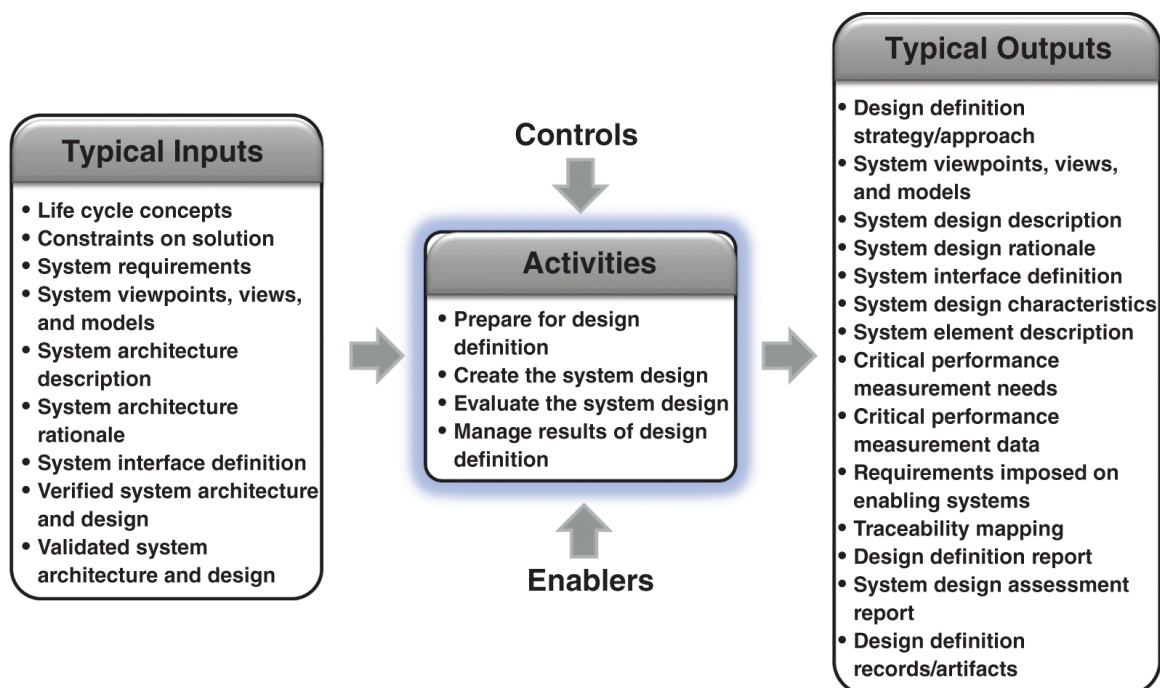


FIGURE 2.44 IPO diagram for Design Definition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Design Definition process includes the following activities:

Prepare for design definition.

- Determine design drivers for the system design and an appropriate design strategy and applicable approaches. Personnel (together with human factors), processes, products, and services intended to compose the system are among the many factors that will impact system design.

Non-functional considerations and design constraints should be identified as these can also serve as design drivers.

- Determine the necessary technologies and the categories of system characteristics to be represented in the design. Capabilities, resources, and services should be identified as these can provide the necessary technologies. Quality models should be identified as these can categorize system characteristics.
- Examine the system architecture to determine the fundamental properties and concepts that apply to the system design, along with the principles that should govern the design and its evolution.
- Establish the approach for system design effort.
- Ensure the necessary system design-enabling elements, services, resources and capabilities are available.

Create the system design.

- Identify and assess design alternatives and create a system design using appropriate techniques such as adapting an existing design, composing a design from available system elements, creating a new design, or through a combination of these approaches. [Section 3.2.7](#) briefly explains several design approaches.
- Allocate system requirements to system elements so that all the system requirements and architecture objectives are addressed. This task includes transformation of architectural entities and relationships to design elements and transformation of architectural characteristics into design characteristics.
- Where necessary, conceive a system design that does not already exist using synthesis techniques, e.g., brainstorming, analogical thinking, or using morphological charts. Synthesis can be applied at multiple levels within a system and focuses upon the solution space and the development of one or more potentially satisfactory design solutions.
- Transform architectural entities (e.g., enterprise or project goals, capabilities and effects, operational activities, resource functions) and relationships into design elements. Also transform architectural characteristics into design characteristics (e.g., functionality, behavior,

dimensions, shapes, materials, critical quality characteristics, data processing structures).

- Analyze the design as part of the design creation activity. A range of analysis techniques can be employed including (1) parametric design to explore within the potential solution space areas for further investigation, and (2) trade-off analysis to achieve balanced system design solutions. Analysis of the design which is systemic in nature (e.g., completeness and consistency analysis) is included in the Systems Analysis process (see [Section 2.3.5.6](#)).
- Analyze the interfaces that were identified and defined in the system architecture and refine or further define the interfaces to the level required for the design characteristics and interactions between the system elements and with external entities.
- Capture the system design expressing it as a work product, termed the system design description, addressing the composition, properties, and characteristics of the system design. Typically, methods, frameworks, diagrammatic forms, notations, and other forms of model are applied to express this work product. The system design description is developed into a specification which can be used to either procure or otherwise realize the system elements which comprise the design.

Evaluate the system design.

- Determine the overall suitability and “goodness” of potential design solutions in meeting the identified need or opportunity and in satisfaction of requirements and constraints. The design should be consistent with a governing architecture description.
- Analyze further the system design in support of such evaluation, for example to determine particular design properties and characteristics.
- Assess the value or worth the design will have for various stakeholders as well as potential negative consequences.
- Combine the analyses and assessments into an overall evaluation determination which can serve as the basis for selecting a preferred system design and serve as feedback to the System Architecture

Definition process. This design evaluation activity can also provide useful information to the Verification process.

Manage results of design definition.

- Establish a means for the management of the system design, including agreement of the design.
- Maintain bidirectional traceability of the system design. Track the satisfaction of requirements, constraints, and objectives, as well as between the design characteristics and architecture entities, interfaces, and analysis results.
- Capture, maintain, and manage the rationale and supporting information for selections among alternatives and decisions about the design. Maintain the design configuration(s) and, in conjunction with the Configuration Management process, conduct change management.
- Manage certification of the design to qualify it as meeting specified quality standards, as applicable (e.g., for critical systems).
- Register the design to protect it and its constituent intellectual property, as applicable.
- Maintain design integrity during the development and evolution of the system design. A specific responsibility (e.g., a system design authority) is generally assigned to address this issue.
- Conduct design reviews to evaluate design progress, suitability, and quality, and record resulting actions and their satisfaction.

Common approaches and tips.

- Identify the enabling systems and materials needed for transition early in the life cycle to allow for the necessary lead time to obtain or access them.
- It is important to maintain good communication and coordination between the SE practitioners and the various practitioners from other disciplines during the design definition of the system elements, in order to ensure a holistic view as the elements evolve.

Elaboration

System Design versus System Architecture The Design Definition process focusses on developing an overall system design that is ultimately sufficiently detailed to allow its realization. It may be driven from stakeholder concerns, requirements, constraints, business opportunity, mission need, or from the architecture developed by the System Architecture Definition process. On the other hand, the system architecture focuses on the essential concepts, properties, structure, behaviors and features that apply to the system solution, the system design ultimately will capture a description of the proposed system solution which is sufficiently detailed to enable its implementation. System design focuses on system technical considerations such as the solution system elements, their interfaces and characteristics together with technological and other realization considerations such as materials, manufacture, software coding, and operator profiles.

Influence of Design Thinking Design Thinking focuses on design processes and reasoning, together with the resulting design concept and their development, specialization and realization (see [Section 3.2.7](#)). It has particular influence upon the creation of the system design. Design Thinking can have as important an influence on system design as Systems Thinking considerations (see [Section 1.5](#)) which focus on taking a holistic, systemic view.

Identification of Design Drivers Design drivers, often identified during the System Architecture Definition process, are those factors which should most heavily influence the system design. They may be identifiable from analysis of the business market for the system, operational considerations, through-life considerations, stakeholder requirements and constraints, human factors, technology characteristics and constraints, implementation factors, etc. The set of design drivers applicable to a particular system problem (and solution) determine the overall design approaches that should be employed, the applicable design principles, and the specific

specialty considerations which should be employed and integrated into a coherent overall system design.

Usually, the functionality required of a system dominates the design considerations. However, in some situations, the non-functional attributes (e.g., safety, security, performance, dependability) will dominate the Design Definition process, for example, in propulsion systems (e.g., gas turbine engines), and for some kinds of systems (e.g., sensor systems) where quality attributes are key. Furthermore, in socio-technical systems, the relationships between system elements and things in the external world (e.g., financial, legal, social, economic, political), as well as human factors considerations, will often dominate the design of such systems. (*For example, see human-centric design*).

Design For X (DFX) Design drivers concerning considerations such as testability (DFT) or manufacturability (DFM), for example, are addressed by quality characteristics described further in [Section 3.1](#). These drivers may apply to either the whole or just a part of the system and a commonly employed system design tactic is to concentrate their applicability just to a specific part of the system. The resulting system design elements or influences need to be integrated into an overall balanced, viable, and satisfactory design using holistic design practices.

Different Approaches used for developing a System Design A system design approach may originate from one or more sources:

- From requirements and constraints as identified in the Stakeholder Needs and Requirements Definition Process, and System Requirements Definition processes
- From the business or mission problem or opportunity and potential solution space characterization as determined by the Business or Mission Analysis process
- From the system architecture as defined by the System Architecture Definition process
- From the Design Strategy (e.g., maximum reuse).

Assurance of a System Design In certain circumstances, system designs are required with high levels of special characteristics such as integrity or dependability and this is likely to require either the more stringent application of design practices or the usage of specific practices (as part of the create the system design activity) to achieve design characteristics such as overall quality or some specific characteristics such as airworthiness, reliability, safety or security. Some of these characteristics may be externally regulated. Specific organizations (authorities) may have been delegated responsibility for identifying appropriate requirements and determining their satisfaction (often by independent means) by a system design. In addition, certification of the system design (as part of the Verification and Validation processes) may be undertaken to ensure that the system design complies with regulatory design certification requirements. This includes matters such as nuclear power plant operational certification, aircraft airworthiness certification, space launch range safety certification, and consumer product safety and emissions certifications.

Notions and Principles Used within Design The Design Definition process identifies the human activities and material system elements for realizing the system solution. Specialist technical disciplines will need to be harnessed and orchestrated to ensure achievement of a coherent overall solution that meets the specified or identified needs or opportunities. Every technical domain or discipline possesses its peculiar laws, rules, theories, and practices for developing solution parts. Designing an overall system entails identifying where and how specific practices should be employed and integrated. Common and coherent design descriptors should be employed across the system to ensure that overall required system properties are realized and that system elements are capable of interoperation.

Usage of Design Descriptors A design descriptor is the set of (1) design characteristics and (2) their possible values. System design entails the identification and quantification of relevant design descriptors for the system elements composing the system design. These descriptors may be

determined through a combination of (1) top-down apportionment and allocation, and (2) bottom-up selection and measurement. Matching system elements together with their specific design descriptors to those of the overall system is a key part of system design.

- The following are examples of generic design characteristics that are specifically relevant to SE:
 - – For overall system functional and structural aspects: Quality of service, modularity, openness, scalability, deployability, and degree of automation and autonomy
 - – For overall system non-functional aspects: Aesthetics, commonality, balance, availability, reliability, affordability, and other relevant QCs (see [Section 3.1](#))
- The following are examples of generic design characteristics that are related to other engineering disciplines, but are relevant to SE in ensuring system balance and supporting trade-offs across system elements:
 - – For hardware elements: Weight/mass, power, geometry, volume, vibration, acoustic, and thermal
 - – For software elements: Correctness, understandability, efficiency, maintainability, flexibility, and consistency
 - – For data/information: Accuracy, completeness, reliability, relevance, and timeliness

Holistic Design System design starts with the system as a whole consisting of system elements and ends with a definition (i.e., design specification) for each of these system elements and how they are designed to work together as a complete system. The system architecture identifies the system elements, although the architecture description might only identify those elements that are architecturally significant with additional elements becoming evident as a result of more detailed design considerations.

Early Design Validation Techniques such as modeling, simulation, and prototyping can be employed to discover early problems in a system design with respect to stakeholder expectations before significant expenditure in

effort and materials. Increasingly these make use of computer-based representations of the system design, including augmented and virtual reality. Such validation techniques can be employed as part of different design process cycles including staged design and as part of an iterative design cycle.

Maintenance of Design Integrity Design integrity concerns the completeness, consistency, and inherent quality characteristics of a system design. An important design responsibility concerns ensuring the integrity of a system design, and subsequently approving, or denying, any decisions concerning the design, its modification, and its implementation. This is important to the acquirer (and therefore also to the supplier) as a system design evolves throughout the Design Definition process, including any evolution following its realization and transition to operation.

Design Evolution The system design matures and evolves over time. Conceptual design focuses on the overall form of the design solution. Preliminary design (sometimes called embodiment) elaborates the system design(s) with more details than provided in the architecture, so that layout, technical, cost, and other realization issues can be addressed. Detailed design progresses the preliminary system design with specification such that the system can be realized unambiguously. Usually, a design continues to evolve for various reasons: to accommodate new technologies, address obsolescence, improve performance and functionality, account for a new threat to its operation and/or integrity, etc. In case the SoI is operated in an SoS environment, there can be continuous design evolution. Consequently, robustness and resilience of the system design need to be considered often and on many levels of the hierarchy. The system architecture can specify principles for evolution of the design. Where possible, features should be included in the design to accommodate change according to the specified architectural principles and design objectives.

2.3.5.6 System Analysis Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.6.1] The purpose of the System Analysis process is to provide a rigorous basis of data and information for technical understanding to aid decision-making and technical assessments across the life cycle.

Description Analysis is “a detailed examination or study of something so as to determine its nature, structure, or essential features” (Oxford, 2020). System analysis uses models and simulations to assess the utility and integrity of system requirements, architecture, and design across the life cycle. Models are incomplete representations of reality that answer four types of questions: (1) descriptive—what has happened? (2) predictive—what could happen? (3) prescriptive—what should we do? And (4) definitive—how should an entity be defined? (Buede and Miller, 2016, p. 70; Lustig, et al., 2010; Watson IOT, 2017).

System models define quantitative (numerical) and qualitative (categorical) variables to represent system attributes along multiple dimensions that can be categorized according to [Figure 2.45](#). By establishing the relationships among the selected variables, the models represent the emergent features of the system and provide answers to the four types of questions from a holistic perspective. In addition to addressing the complexity of many interacting variables, modeling approaches may be needed to address uncertainty, dynamic behavior, and feedback loops (Howard, 1968). Typical modeling approaches include MBSE with functional, structural, and behavioral modeling (Dennis, et al., 2020); mathematical analysis; probabilistic and statistical modeling; simulation; and other techniques to represent the relationships among the variables and to perform sensitivity analysis of the allowable range of values for the variables across all life cycle stages. In some cases, it may be necessary to employ a mixture of modeling approaches to obtain the necessary insight. For more information on models and simulation, see [Section 3.2.1](#).

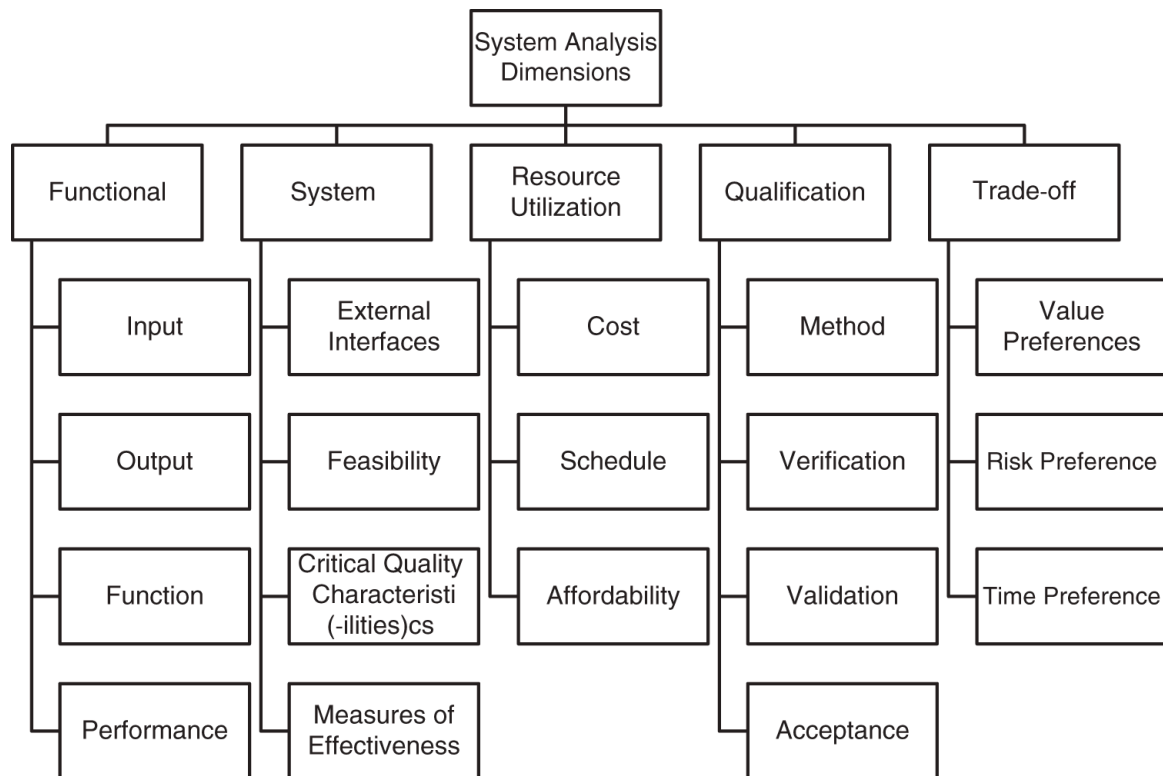


FIGURE 2.45 Taxonomy of system analysis dimensions. INCOSE SEH original figure created by Kenley. Usage per the INCOSE Notices page. All other rights reserved.

Inputs and Outputs Inputs and outputs for the System Analysis process are listed in [Figure 2.46](#). Descriptions of each input and output are provided in Appendix E.

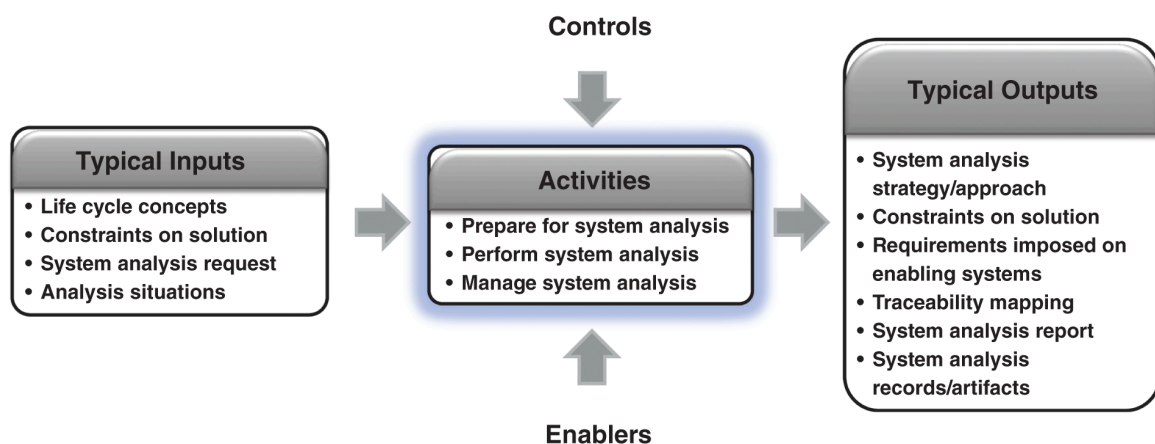


FIGURE 2.46 IPO diagram for System Analysis process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The System Analysis process includes the following activities:

- *Prepare for system analysis.*

- – Establish the strategy/approach for system analysis.
- – Identify the situation (problem, opportunity, question, or decision) to be addressed by system analysis.
- – Identify the stakeholders and their perspectives for the system analysis.
- – Define the scope (including system level), objectives, and level of fidelity of the system analysis.
- – Choose analysis methods based on relevant drivers, such as time, cost, fidelity, and criticality.
- – Plan for the necessary enabling systems or services needed through the life cycle for system analysis.
- – Ensure enabling system or service access needed to support system analysis.
- – Identify assumptions and ensure they are valid (correct and consistent).
- – Ensure the data and inputs needed for the analysis are collected are timely and trustworthy.
- – Establish criteria for trustworthiness of data needed for the analysis.
- – Review the data and inputs for quality and validity (i.e., trustworthy data).
- *Perform system analysis.*
 - – Execute the required system analysis using the selected analysis methods.
 - – Perform uncertainty analysis and sensitivity analysis
 - – Perform a quality and validity check on the analysis results.
 - – Interpret the analysis results to develop relevant conclusions and recommendations.
 - – Capture the system analysis results. This is often done in a system analysis report.
- *Manage system analysis.*
 - – Establish and sustain traceability (system analysis results).
 - – Give CM the information items, work products, or other artifacts needed for baselines.

Common approaches and tips:

- Models can never simulate all the behavior of a system: they operate in limited fields with a restricted number of variables. When a model is used, it is always necessary to make sure that the parameters and data inputs are part of the operation field; if not, irregular outputs are likely.
- Models evolve during the project: by modification of parameters, by entering new data, and by the use of new tools.
- It is recommended to concurrently use several types of models in order to compare the results and to take into account another characteristic or property of the system.
- Results of a simulation shall always be given in their modeling context: tool used, selected assumptions, parameters and data introduced, and variance of the outputs.

Elaboration

System Analysis Relationships to Other System Life Cycle Processes. Some of the SE processes leverage different dimensions of system analysis to answers different questions are as follows:

- The Business or Mission Analysis process to analyze and estimate candidate OpsCon and/or candidate business models related to a potential SoI in terms of effectiveness, feasibility, costs, risk preferences, and value preferences that address uncertainty, dynamic behavior, and feedback based on operator or end user behavior (Choi, et al., 2020).
- The Stakeholder Needs and Requirements Definition and System Requirements Definition processes to analyze issues relating to conflicts among the set of requirements, in particular those related to functionality, feasibility, performance, and effectiveness, including technical risks, costs, and externally imposed operational conditions and constraints (Beery and Paulo, 2019).
- The System Architecture Definition and Design Definition processes to analyze and estimate architectural and design space characteristics of candidate architectures and/or system element, providing information

needed to select a design that provides the best value in terms of feasibility, and effectiveness, including technical risks, costs, and critical quality characteristics such as dependability, affordability, maintenance, and human-system interface considerations (Guariniello, et al., 2020).

- The Verification and Validation processes to understand and quantify the cost, schedule, information value, and the uncertainty characteristics inherent among the different choices of methods among inspection, analysis (including simulation), demonstration, and test (Salado and Kannan, 2018).
- The Project Planning and Project Assessment and Control processes to obtain estimates along with range of uncertainty of system metrics against established targets and thresholds, especially with respect to the technical measures (MOEs, MOPs, and TPMs) (Raz, et al., 2020).

The results of system analyses and estimations are provided to the Decision Management process as data, information, and arguments for selecting the alternatives or candidates that provide the best value to the decision maker(s) based on their value, risk, and time preferences. In some cases, the results may be provided to the Project Assessment and Control process, if the information is needed to monitor the progress of the system or project against its system objectives, performance thresholds, or growth targets.

Cost Analysis. A cost analysis should consider the life cycle costs (LCC), which can be adapted according to the project and the system. The LCC may include labor and non-labor cost items; it may include development, manufacturing, service realization, sales, stakeholder utilization, supply chain, maintenance, and disposal costs (see [Section 3.1.2](#)).

Technical Risk Analysis. Technical risks address the operational system as opposed to the project risks, which are concerned with developing the system. The System Analysis process is often needed to perform the technical assessments that provide quantification and understanding of the probability or impact of a potential risk or opportunity (see [Section 2.3.4.4](#)).

Effectiveness Analysis. System effectiveness analysis is a term for a broad category of analyses that evaluate the degree or extent to which a system meets one or more criteria in its intended operational environment. The criteria may be derived from desired system characteristics, such as TPMs, MOPs, MOEs, or other attributes of the system (see [Section 2.3.4.7](#)). The system analysis assesses the probability the criteria are met and also the degree to which they are met (or fall short or exceed). This information is used to support evaluation of alternatives and trade-offs (such as which candidates to develop further, where improvements are needed or cost savings are possible).

2.3.5.7 Implementation Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.7.1] The purpose of the Implementation process is to realize a specified system element.

The Implementation process creates the system element per that element's description (concepts, requirements, architecture, design, including interfaces). Note that this process does not only occur during the production stage of the life cycle, but has activities in the other stages to ensure the element can be produced and to prepare for the production stage or other stages. For example: the supporting infrastructure may need to be defined or upgraded in preparation for the activities in a stage.

Description During the Implementation process, engineers follow the requirements allocated and derived to the system element to fabricate, code, or build each individual element outlined in system element descriptions. System element requirements are verified and system element stakeholder requirements are validated. If subsequent configuration audits reveal discrepancies, iterative and recursive

interactions occur with predecessor activities or processes, as required, to correct them.

Inputs/Outputs Inputs and outputs for the Implementation process are listed in [Figure 2.47](#). Descriptions of each input and output are provided in Appendix E.

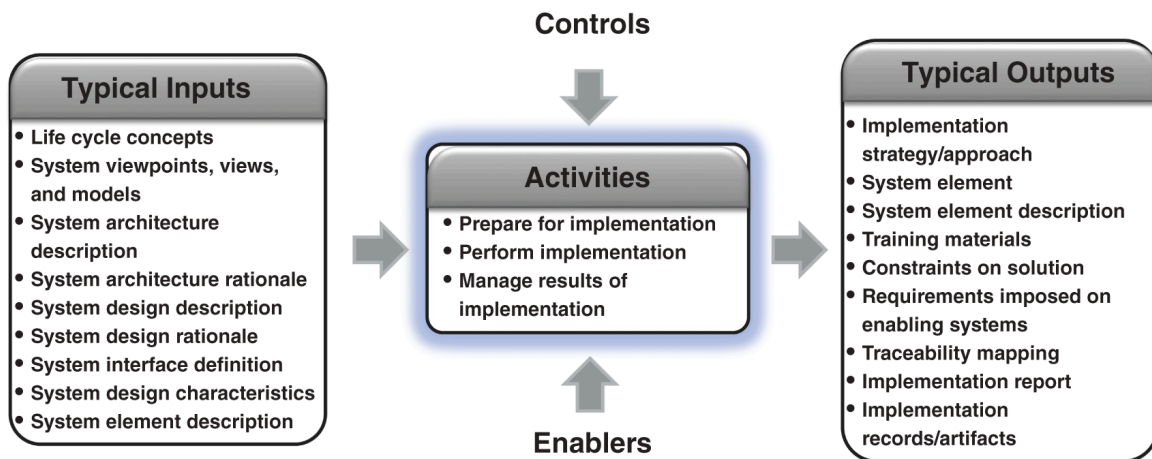


FIGURE 2.47 IPO diagram for Implementation process. IISE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Implementation process activities include the following:

- *Prepare for implementation.*
 - – Define fabrication/coding procedures, tools and equipment to be used, implementation tolerances, and the means and criteria for auditing configuration of the resulting elements.
 - – Elicit from stakeholders, developers, and teammates any constraints imposed by implementation technology, strategy, or impacted systems. Record the constraints for consideration in the definition of the requirements, architecture, design, and implementation.
 - – Document the plan for acquiring or gaining access to resources needed during implementation. The planning includes the identification of requirements and interfaces for the enabling system.